

POST-TROPICAL SCREENING

CASE HISTORIES

Version 2004

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You are commissioned with the medical care of a unit that has been deployed during 3 months in a humanitarian operation (involving medical relief and logistics) among refugees in an East African country.

The unit included 40 male and female personnel. All had been immunised against diphtheria, tetanus, poliomyelitis, yellow fever, hepatitis A and hepatitis B prior to departure, and advised to take mefloquine 250 mg weekly for malaria chemoprophylaxis. The camp consisted of tents, had its own kitchen and sanitary facilities, and was located on the side of a lake that is known to be endemic for *Schistosoma haematobium* and *S. mansoni*. It was rainy season for most of the deployment period.

Post-tropical screenings are to be done for the entire unit. The commanding officer prefers prompt diagnostic action, since most of the unit have been in contact with lake water, and recently two cases of Katayama syndrome occurred in aid workers who had been swimming there. His personnel are also keen on speedy investigations, as they want to be "worm-free" as soon as possible. For practical reasons your superiors have planned the screening to coincide with the debriefing, to be held in the first week after return.

Question 1

What is your opinion on the proposed timing of the post-tropical screening?

Your superiors are worried about the malaria risk. They want the unit to be checked for malaria as soon as possible.

Question 2

Is this useful? If it is, which test should be used?

Question 3

Would there be reason to screen the unit at any time for schistosomiasis? Which diagnostic method would you use?

Question 4

Is it useful to test for schistosomiasis shortly after return to prevent the Katayama syndrome? Why (not)?

Upon first screening, 3 days after return, 6 personnel have abdominal symptoms.

For 2 of these, the symptoms had started on the day of return, and include nausea, watery diarrhoea without blood, cramps and fever up to 38.5° C. One had vomited.

The remaining 4 have had symptoms since 2 to 5 weeks, including epigastric pain, loss of appetite and 2 to 3 loose, greasy stools daily. One lost 3 kg weight over one month. There has been no fever. Two had an episode of acute, watery diarrhoea, 5 and 8 weeks earlier, respectively.

Question 5

Which elements in the medical history do you find most important for making your differential diagnosis?

Question 6

What are the most likely diagnoses for the 2 patients with diarrhoea of recent onset?

Question 7

What are the most likely diagnoses for the remaining 4 patients? What would be your diagnostic plan?

Two of these 4 turn out to have eosinophilia, with 600 and 1950 eosinophils/microliter, respectively.

Question 8

Which diagnoses would you consider in these cases? What would you like to know regarding specific exposures?

Question 9

What would be your diagnostic plan for these 2 patients?

Several other personnel reported episodes of diarrhoea without fever or blood loss during the mission. One had a stool test in a local lab, and was diagnosed with amoebiasis. He was treated successfully with metronidazol for 10 days. He is currently without symptoms, but his stool test result now reads ‘cysts of *Entamoeba histolytica*’.

Question 10

What is your diagnosis?

Question 11

How do you explain this finding?

Question 12

Would you treat this patient, and if yes, how?

The protocol suggests screening for latent tuberculosis infection by either chest X-ray or tuberculin skin testing for all returned personnel.

Question 13

Which TB screening method would you prefer:

- a. in personnel *without* a history of BCG vaccination?
- b. in personnel *with* a history of BCG vaccination?
- c. in personnel *with* a history of BCG vaccination but a negative tuberculin test after vaccination?

Question 14

How would you estimate (in general terms) the cost-effectiveness of chest X-ray screening in this group?